Vitor Alexandre Garcia Vaz

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Summary

Computer Engineering student at USP São Carlos. Develops hardware and software together: schematics and PCBs, real-time RTOS firmware on microcontrollers, sensor drivers, digital filtering (with emphasis on frequency filters and the Kalman filter) and long-range wireless communication. Systems validated in sounding rocket flights and in low-power IoT network research.

Education

University of São Paulo (USP) — EESC and ICMC São Carlos, SP
Bachelor's in Computer Engineering *2023 - 2027 (expected)*

Experience

Avionics Coordinator TOPUS Aerospace Projects — EESC-USP 05/2025 - 09/2026
- Coordinated development of the Atlas flight computer, ensuring each subsystem (communication, sensing, power, deployment) worked reliably and in isolation
- Wrote the C++/FreeRTOS firmware that keeps the flight intact even after an airborne reboot, with sensor fusion (Kalman and Madgwick), critical data persistence and 99.89% apogee accuracy against a commercial altimeter
- Contributed to the C++ firmware and KiCad PCBs of four of the group's vehicles, including the 2U CubeSat awarded 2nd place at LASC 2026
- Standardized support for multiple LoRa radio models into a single C++ library, eliminating code duplication across the group's projects

Undergraduate Research LaSDPC — ICMC-USP 04/2025 - Present
- Developed C++/FreeRTOS firmware for a LoRaWAN node, using deep sleep to extend field battery life
- Designed the node's own PCB in KiCad, integrating radio, sensors and battery into a compact field platform
- Built the field testbed that let the team measure range and power consumption of each protocol

Embedded Systems Department Director Zenith Aerospace — EESC-USP 03/2024 - 06/2025
- As Director of the Embedded Systems Department, trained new members and managed the subsystem
- Resolved C driver duplication across the group's platforms, unifying them into a C hardware abstraction layer portable across different microcontrollers (such as STM32, Arduino and ESP32)

Project Manager SAEComp — USP 05/2025 - 03/2026
- Gathered requirements, designed the PostgreSQL database and led the department office's online store team; the store increased the office's profit 75-fold
- Developed the store's Node.js/TypeScript REST API, eliminating manual PIX payment reconciliation

Projects

Flex Ground Station | *KiCad, ESP32, C++, FreeRTOS, LoRa, I2C/SPI* 05/2026 - 09/2026
- Portable ground station with its own KiCad PCB (ESP32, RFM95W, OLED, battery) and C++/FreeRTOS firmware with 5 core-pinned tasks, with field-adjustable radio (RSSI/SNR)

LoRaWAN Sensor Node | *ESP32, C++17, KiCad, FreeRTOS, MCCI LMIC* 01/2026 - 09/2026
- C++/FreeRTOS firmware with LoRaWAN radio (AU915) and deep sleep for power savings, with its own PCB in KiCad integrating ESP32, sensors and battery

LoRa Communication Library | *C++, PlatformIO* 10/2025 - 09/2026
- C++ library that abstracts 4 LoRa radios behind compile-time flags, used across the whole group

Technical Skills

Languages: C, C++, Python, TypeScript, Java
Microcontrollers & RTOS: ESP32, STM32 (HAL), Arduino, FreeRTOS, PlatformIO, STM32CubeIDE
Protocols: I2C, SPI, UART, LoRa, LoRaWAN, MQTT
Sensors: IMU (MPU6050, LSM6DS3), barometers (BMP280, BMP390), GNSS, INA219, DHT22
Hardware: KiCad (schematic, layout, libraries), PCB manufacturing rules
Tools: MATLAB, Simulink, Git, Linux, Docker, LaTeX | **Spoken Languages**: Portuguese (native), English (advanced)

Awards

Best Paper Award, IEEE SmartIoT 2026, as co-author of a paper on a reproducible, observable IoT testbed
2nd place in the CubeSat Category, Latin American Space Challenge (LASC) 2026, with the Atena CubeSat (TOPUS)